

AI Product Management Certification

by Product Faculty

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Complete Course Notes — All 9 Modules

From Identifying AI Opportunities to Landing the AI PM Role

Course Overview

Module	Title
Module 1	Identify High-Leverage AI Opportunities Using the 4D Method
Module 2	Design Trustworthy AI Experiences That Drive Adoption
Module 3	Systematically Optimize Outputs for Accuracy & Relevance
Module 4	Build Rigorous Evaluation Suites — Not Vibes
Module 5	Architect Agentic AI Systems That Work
Module 6	Select the Perfect LLM for Cost, Latency & Quality
Module 7	Navigate AI Ethics, Safety & Responsible Deployment
Module 8	Lead AI Teams, Manage Stakeholders & Drive Adoption
Module 9	Build Your AI PM Portfolio & Land the Role

Module 1: Identify High-Leverage AI Opportunities Using the 4D Method

Lesson 1.1 — Start With User Pain, Not Technology

Rohan speaking:

The #1 mistake AI PMs make is starting with the technology. "We have access to GPT-4, what should we build?" This is backwards. The right question is: "What is the most painful thing our users do today, and can AI solve it better than anything else?"

The AI Opportunity Stack

```
LAYER 1: User Pain (Start here)
  What is the most frequent, high-severity problem users face?

LAYER 2: Current Solution Gap
  Why do existing solutions fail? What's the workaround?

LAYER 3: AI Suitability
  Is this a pattern recognition problem? Does it require language understanding?
  Is there enough data? Is the cost of being wrong acceptable?

LAYER 4: Business Value
  If AI solves this, what's the measurable business impact?
```

Lesson 1.2 — The 4D Method

Henry speaking:

The 4D Method is your systematic framework for finding where AI creates real value.

D	Step	Key Question
Discover	Map all user workflows	Where do users spend the most time? Where do they make the most errors?
Define	Identify the highest-pain workflow	Which workflow, if improved, would have the biggest impact on user success?
Determine	Apply AI Suitability Test	Is AI the right solution? Or is this a process/UX problem?
Decide	Choose the AI approach	RAG, Agent, Classifier, or Generator — which fits the problem?

Lesson 1.3 — The 5-Question Framework to Quantify Pain

Rohan speaking:

Question	What It Measures	Score (1–5)
Frequency	How often does this pain occur?	Daily=5, Weekly=3, Monthly=1
Severity	How bad is it when it happens?	Blocks work=5, Slows work=3, Minor=1
Population	How many users experience this?	>80%=5, 40-80%=3, <40%=1
Workaround	How painful is the current workaround?	No workaround=5, Bad workaround=3, OK workaround=1
Unlocks	What does solving this enable?	New revenue=5, Retention=3, Efficiency=1

Pain Score Formula: (Frequency + Severity + Population + Workaround + Unlocks) / 25 × 100

Score >70: Strong AI opportunity. Score 40–70: Investigate further. Score <40: Probably not worth it.

Lesson 1.4 — The AI PRD: What's Different

Henry speaking:

A standard PRD has problem statement, user stories, success metrics, and out of scope. An AI PRD has all of that PLUS 7 additional sections that most PMs skip — and that's why most AI products fail.

- **AI Approach:** RAG / Agent / Classifier / Generator — and why this approach
- **Data Requirements:** What data does the model need? What's the quality bar?
- **Model Selection Rationale:** Which model tier and why (cost vs. quality trade-off)
- **Prompt Strategy:** High-level prompt architecture and iteration plan
- **Evaluation Plan:** Golden Dataset size, judge rubric, key metrics
- **Guardrails Specification:** Input guardrails, output guardrails, human oversight
- **Cost Model:** Tokens × price × volume = monthly cost estimate
- **AI Risk Register:** Top 5 risks with probability, severity, and mitigation

Module 1 Lab Assignment

- Build an AI Opportunity Assessment using the 4D Method
- Apply the 5-Question Framework to score your opportunity
- Write a 1-page AI PRD with all AI-specific sections

Module 2: Design Trustworthy AI Experiences That Drive Adoption

Lesson 2.1 — The #1 UX Mistake in AI Products

Rohan speaking:

The #1 UX mistake is adding a chat interface and calling it an AI product. Chat is one interaction pattern. It's not always the right one. The right interaction pattern depends on the user's workflow, context, and cognitive load.

AI Interaction Pattern Matrix

Pattern	Best For	Example
Chat / Conversational	Open-ended exploration, Q&A, support	Customer support bot, research assistant
Inline Suggestion	Augmenting existing workflows	GitHub Copilot, Gmail Smart Compose
Proactive Nudge	Surfacing insights users didn't ask for	Slack summarization, CRM next-best-action
Automated Action	Repetitive, low-stakes tasks	Auto-categorization, auto-tagging
Structured Output	Generating documents, reports, summaries	PRD generator, meeting notes

Lesson 2.2 — The 3P Framework for AI UX

Henry speaking:

For every AI feature, evaluate it on three dimensions:

- **Prioritization:** Is this AI feature solving the highest-priority user pain? Or is it a nice-to-have?
- **Placement:** Is the AI feature placed where users are already working? Or does it require them to go somewhere new?
- **Prominence:** Is the AI feature visible enough to be discovered? Or is it buried in a settings menu?

Lesson 2.3 — Designing for Trust

Rohan speaking:

Trust is the #1 adoption driver for AI features. Users who don't trust the AI won't use it — even if it's accurate.

The 5 Trust Signals in AI UX

Trust Signal	How to Implement It
Source Transparency	Show where the AI got its information. "Based on your last 3 meetings..."
Confidence Indicators	Show when the AI is uncertain. "I'm not sure about this — you may want to verify."
Human Override	Always give users a way to edit, reject, or escalate AI outputs.
Explainability	Show the reasoning, not just the answer. "I recommended this because..."
Error Recovery	Make it easy to report bad outputs and get a human alternative.

Lesson 2.4 — Designing AI Error States

Henry speaking:

AI fails differently from traditional software. Traditional software either works or crashes. AI degrades gracefully — or ungracefully, if you haven't designed for it.

The 4 AI Error States

Error State	What Happens	Design Response
Hallucination	AI generates confident but wrong information	Source citations, confidence scores, human review flag
Refusal	AI declines to answer (over-cautious guardrails)	Clear explanation + alternative path (human handoff)
Irrelevance	AI answers a different question than asked	Clarification prompt, "Did you mean...?" pattern
Timeout/ Failure	Model API is slow or unavailable	Graceful degradation, cached responses, human fallback

Module 2 Lab Assignment

- Map the interaction patterns for your AI feature using the matrix
- Apply the 3P Framework — score your feature on Prioritization, Placement, Prominence
- Design the 5 trust signals for your specific feature
- Design all 4 error states with specific UX responses

Module 3: Systematically Optimize Outputs for Accuracy & Relevance

Lesson 3.1 — Why Output Quality Is a PM Problem

Henry speaking:

Most PMs treat output quality as an engineering problem. "The model will figure it out." This is wrong. Output quality is determined by three things: the model you choose, the data you provide, and the prompt you write. PMs own two of those three.

Lesson 3.2 — The Prompt Engineering Hierarchy

Rohan speaking:

Level	Technique	When to Use	Quality Gain
1	Zero-Shot	Simple, well-defined tasks	Baseline
2	Few-Shot	Tasks with specific output format	+15–25%
3	Chain-of-Thought	Reasoning tasks, multi-step problems	+20–35%
4	Role + Context + Constraints	Domain-specific tasks	+25–40%
5	RAG (Retrieval-Augmented)	Tasks requiring specific knowledge	+40–60%

Lesson 3.3 — The System Prompt Architecture

Henry speaking:

A production system prompt has 6 components. Missing any one of them degrades quality.

```
COMPONENT 1: ROLE
"You are [specific role] at [specific context]."
Example: "You are a senior product manager at a B2B SaaS company."

COMPONENT 2: TASK
"Your job is to [specific task] given [specific inputs]."
Example: "Your job is to write user stories given a feature description."

COMPONENT 3: CONTEXT
"Here is the relevant context: [context]"
Example: "Here is our product documentation: [docs]"

COMPONENT 4: CONSTRAINTS
"You must always [constraint]. You must never [constraint]."
Example: "You must always include acceptance criteria. Never exceed 3 user stories."

COMPONENT 5: OUTPUT FORMAT
```

```
"Format your response as [format]."  
Example: "Format your response as: As a [user], I want [action] so that [benefit]."  
  
COMPONENT 6: EXAMPLES (Few-Shot)  
"Here are examples of good outputs: [examples]"
```

Lesson 3.4 — RAG Architecture for PMs

Rohan speaking:

RAG (Retrieval-Augmented Generation) is the most important AI architecture pattern for product PMs to understand. It solves the #1 problem with LLMs: they don't know your specific data.

How RAG Works (Plain English)

```
USER QUERY  
  |  
  ▼  
RETRIEVAL ENGINE  
(Search your knowledge base for relevant chunks)  
  |  
  ▼  
CONTEXT ASSEMBLY  
(Combine retrieved chunks with the user query)  
  |  
  ▼  
LLM  
(Generate answer grounded in retrieved context)  
  |  
  ▼  
GROUNDED RESPONSE  
(Answer based on your data, not model training data)
```

The 4 RAG Quality Levers

Lever	What It Controls	PM's Role
Chunking Strategy	How documents are split for retrieval	Define chunk size requirements based on use case
Retrieval Quality	How well the search finds relevant chunks	Define relevance criteria, test with real queries
Context Window	How much context is passed to the model	Balance quality vs. cost (more context = more tokens)
Grounding Instructions	How strictly the model sticks to retrieved context	Define faithfulness requirements in system prompt

Module 3 Lab Assignment

- Write a production-quality system prompt using all 6 components

- Test your prompt at 3 levels of the hierarchy — measure quality difference
- Design a RAG architecture for your AI feature
- Define your 4 RAG quality levers with specific requirements

Module 4: Build Rigorous Evaluation Suites — Not Vibes

Lesson 4.1 — Why Evaluation Is the Most Important PM Skill in AI

Henry speaking:

"If you can't measure it, you can't improve it. In AI, if you can't evaluate it, you're flying blind. Every prompt change, every model upgrade, every architecture decision — you need an eval suite to know if you made things better or worse."

Lesson 4.2 — The Golden Dataset

Rohan speaking:

The Golden Dataset is your ground truth. It's a curated set of inputs with known-good outputs that you use to measure your AI system's quality.

Golden Dataset Composition

Category	% of Dataset	What It Tests
Happy Path	60%	Standard use cases your AI should handle well
Edge Cases	25%	Unusual inputs, ambiguous queries, boundary conditions
Adversarial	15%	Inputs designed to break your AI — jailbreaks, contradictions, trick questions

Minimum viable Golden Dataset: 30 examples. **Production Golden Dataset:** 200+ examples.

Lesson 4.3 — The Evaluation Metrics Framework

Henry speaking:

Metric	What It Measures	How to Score	Target
Faithfulness	Is the output grounded in the provided context?	LLM-as-Judge (0–1)	>0.85
Relevance	Does the output answer the actual question?	LLM-as-Judge (0–1)	>0.80
Completeness	Does the output cover all required elements?	Rubric checklist	>0.75
Latency	How fast does the system respond?	p50, p95, p99 (seconds)	p95 < 3s
Cost per Query	Token cost per user interaction	\$ per 1000 queries	Defined by business case

Lesson 4.4 — LLM-as-Judge

Rohan speaking:

LLM-as-Judge is the technique of using a separate LLM to evaluate the outputs of your AI system. It scales qualitative evaluation that would otherwise require human reviewers.

LLM-as-Judge Prompt Template

```
You are an expert evaluator for an AI [use case] system.

Given the following:
- User Query: {query}
- Retrieved Context: {context}
- AI Response: {response}

Evaluate the response on the following dimensions:

1. FAITHFULNESS (0-1): Is every claim in the response supported
   by the retrieved context? 1.0 = fully grounded, 0 = hallucinated.

2. RELEVANCE (0-1): Does the response directly answer the user's query?
   1.0 = perfectly relevant, 0 = completely off-topic.

3. COMPLETENESS (0-1): Does the response cover all important aspects
   of the query? 1.0 = comprehensive, 0 = missing key information.

Return your evaluation as JSON:
{
  "faithfulness": [score],
  "relevance": [score],
  "completeness": [score],
  "overall": [average],
  "reasoning": "[1-2 sentence explanation]",
  "failure_mode": "[if overall < 0.7, describe the primary failure]"
}
```

Module 4 Lab Assignment

- Build a 30-example Golden Dataset for your AI feature
- Write a custom LLM-as-Judge prompt for your use case
- Run your Golden Dataset through your AI system and score every output
- Identify your top 3 failure patterns and root causes
- Make one improvement and re-run — document the delta

Module 5: Architect Agentic AI Systems That Work

Lesson 5.1 — What Is an AI Agent?

Henry speaking:

An AI agent is an LLM that can take actions in the world — not just generate text. It has three components: Sense (perceive inputs), Plan (decide what to do), Act (use tools to do it).

SENSE LAYER (Inputs)	PLAN LAYER (Reasoning)	ACT LAYER (Tools)
User message Documents API responses Tool outputs	→ LLM decides what to do and in what order	→ Search the web Query a database Send an email Call an API Write a file Execute code

Lesson 5.2 — Agent Planning Patterns

Rohan speaking:

Pattern	How It Works	Best For	Risk
ReAct	Reason → Act → Observe → Repeat	Research, investigation tasks	Can loop indefinitely
Plan-Execute	Create full plan first, then execute steps	Complex multi-step tasks	Plan may be wrong; hard to adapt
Manager-Worker	Orchestrator agent delegates to specialist agents	Complex workflows with distinct subtasks	Coordination overhead, error propagation

Lesson 5.3 — Tool Design for Agents

Henry speaking:

Every tool your agent uses needs a schema. The schema is how the agent knows what the tool does, when to use it, and how to call it correctly.

Tool Schema Template

<pre>{ "name": "search_knowledge_base", "description": "Search the product knowledge base for relevant information. Use this when the user asks about product features, pricing, or policies. Do NOT use for general web search.", "parameters": {</pre>
--

```

    "query": {
      "type": "string",
      "description": "The search query. Be specific. Use keywords from the user's question."
    },
    "max_results": {
      "type": "integer",
      "description": "Maximum number of results to return. Default: 3. Max: 10."
    }
  },
  "returns": "Array of {title, content, source_url} objects",
  "constraints": [
    "Only search the internal knowledge base, not the web",
    "Always use at least 2 keywords in the query"
  ]
}

```

Lesson 5.4 — The 5 Agent Failure Modes

Rohan speaking:

Failure Mode	What Happens	Prevention
Infinite Loop	Agent keeps calling tools without making progress	Max steps limit (e.g., 15 steps), loop detection
Tool Misuse	Agent calls the wrong tool or with wrong parameters	Clear tool descriptions, parameter validation
Context Loss	Agent forgets earlier steps in long tasks	Explicit state object, summarization at checkpoints
Hallucinated Actions	Agent invents tool calls that don't exist	Strict tool list in system prompt, output validation
Scope Creep	Agent takes actions beyond its intended scope	Explicit action constraints, human approval for high-stakes actions

Lesson 5.5 — Agent Reliability Checklist

TERMINATION CONDITIONS:

- ☐ Maximum steps defined (e.g., 15 steps)
- ☐ Maximum duration defined (e.g., 60 seconds)
- ☐ Success condition defined (how does the agent know it's done?)
- ☐ Failure condition defined (when should the agent give up?)

HUMAN OVERSIGHT:

- ☐ Human approval required for irreversible actions
- ☐ Human handoff path for tasks the agent can't complete
- ☐ Audit log of all agent actions

ERROR HANDLING:

- ☐ Retry logic for transient tool failures
- ☐ Graceful degradation if a tool is unavailable

☐ Clear error message to user if agent fails

SAFETY:

- ☐ Agent cannot take actions outside defined scope
- ☐ PII is not logged in agent action history
- ☐ Rate limiting on expensive tool calls

Module 5 Lab Assignment

- Choose a planning pattern for your agent and justify the choice
- Write tool schemas for all tools your agent needs
- Define your agent's state object (full JSON schema)
- Complete the Agent Reliability Checklist
- Identify which of the 5 failure modes your agent is most vulnerable to and define mitigation

Module 6: Select the Perfect LLM for Cost, Latency & Quality

Lesson 6.1 — The Model Selection Framework

Henry speaking:

Model selection is one of the highest-leverage decisions an AI PM makes. The wrong model costs you 10× more than necessary, or delivers 30% worse quality. Here's how to make the decision systematically.

The 3-Dimension Model Selection Matrix

Dimension	Questions to Ask	Drives Toward
Quality Requirements	What's the minimum acceptable quality? What's the cost of a wrong answer?	Higher quality = larger model
Latency Requirements	What's the maximum acceptable response time? Is this real-time or async?	Lower latency = smaller/faster model
Cost Constraints	What's the budget per query? What's the expected query volume?	Lower cost = smaller model or caching

Lesson 6.2 — The Model Tier Framework

Rohan speaking:

Tier	Models	Best For	Cost	Latency
Frontier	GPT-4o, Claude 3.5 Sonnet, Gemini 1.5 Pro	Complex reasoning, high-stakes outputs	High (\$\$\$)	Medium (2–5s)
Mid-Tier	GPT-4o-mini, Claude 3 Haiku, Gemini Flash	Most production use cases	Medium (\$\$)	Fast (0.5–2s)
Small/Fast	GPT-3.5-turbo, Claude Instant	Simple classification, routing, formatting	Low (\$)	Very fast (<0.5s)
Open Source	Llama 3, Mistral, Phi-3	Privacy-sensitive, on-premise, fine-tuning	Infra cost only	Depends on infra

Lesson 6.3 — The Cost Model Every AI PM Must Build

Henry speaking:

MONTHLY COST FORMULA:

Monthly Cost =

```
(Avg Input Tokens per Query × Input Price per 1M tokens / 1,000,000)
+ (Avg Output Tokens per Query × Output Price per 1M tokens / 1,000,000)
× Monthly Query Volume
```

EXAMPLE:

Input: 2,000 tokens × \$3/1M = \$0.006 per query
Output: 500 tokens × \$15/1M = \$0.0075 per query
Total per query: \$0.0135
Monthly volume: 100,000 queries
Monthly cost: \$1,350

COST OPTIMIZATION LEVERS:

1. Prompt compression (reduce input tokens by 20–30%)
2. Response length constraints (reduce output tokens)
3. Caching (cache identical queries – can reduce cost 40–60%)
4. Model routing (use cheap model for simple queries, expensive for complex)
5. Batching (batch async requests for 50% cost reduction)

Lesson 6.4 — Fine-Tuning vs. Prompt Engineering vs. RAG

Rohan speaking:

Approach	When to Use	Cost	Time to Value
Prompt Engineering	Always try first. Solves 80% of quality problems.	Low	Hours
RAG	When the model needs your specific data/knowledge	Medium	Days–Weeks
Fine-Tuning	When you need a specific style, format, or domain expertise that prompting can't achieve	High	Weeks–Months

"Fine-tuning is not a quality shortcut. It's a specialization tool. If you're fine-tuning because your prompts aren't working, you're solving the wrong problem." — Henry

Module 6 Lab Assignment

- Score your AI feature on the 3-dimension matrix (Quality, Latency, Cost)
- Select a model tier and justify your choice
- Build your cost model — calculate monthly cost at 3 volume scenarios (10K, 100K, 1M queries)
- Identify your top 2 cost optimization levers and estimate savings
- Decide: Prompt Engineering, RAG, or Fine-Tuning — and document your reasoning

Module 7: Navigate AI Ethics, Safety & Responsible Deployment

Lesson 7.1 — Why Ethics Is a PM Problem, Not a Legal Problem

Henry speaking:

"At Anthropic, we don't have a separate ethics team that reviews products. Every PM is responsible for the ethical dimensions of their product. That's the only model that works at scale."

Ethics as product quality means: ethical requirements are defined in the PRD, ethics constraints shape architecture from day one, guardrails are native to the UX (not bolted on), and legal review is fast because most issues are already addressed.

Lesson 7.2 — The 6 Ethical Dimensions Every AI PM Must Evaluate

Rohan speaking:

Dimension 1: Fairness & Bias

Test	How to Run It	Red Flag
Demographic parity	Run Golden Dataset with inputs representing different demographics	>10% quality gap between groups
Language parity	Test in all supported languages	>15% quality gap vs. English
Edge case representation	Check if Golden Dataset includes minority use cases	<10% of dataset represents edge cases

Dimension 2: Privacy & Data Governance — The AI Privacy Checklist

DATA COLLECTION:

- ☐ What user data does the AI receive as input?
- ☐ Is any of this data PII? (names, emails, health info, location)
- ☐ Does the user know their data is being processed by AI?
- ☐ Have users consented to AI processing of their data?

DATA STORAGE:

- ☐ Are conversation logs stored? For how long?
- ☐ Are conversation logs used for model training?
- ☐ Can users request deletion of their conversation history?

DATA TRANSMISSION:

- ☐ Is data encrypted in transit to the model provider?
- ☐ Does your model provider's DPA meet compliance requirements?
- ☐ Is any data sent to third-party tools or APIs?

Dimension 3: Transparency & Explainability

When full transparency is required (non-negotiable): AI makes decisions affecting user rights (loans, hiring, insurance), AI generates content that could be mistaken for human-created, AI is used in healthcare/legal/financial contexts, regulations require it (EU AI Act).

Dimension 4: Safety & Harm Prevention — 3 Lines of Defense

```
LINE 1: Input Guardrails
- Content classification on user input
- PII detection and redaction
- Prompt injection detection
- Rate limiting on suspicious patterns

LINE 2: Model-Level Safety
- System prompt constraints
- Negative instructions ("never provide...")
- Safe messaging guidelines for sensitive topics

LINE 3: Output Guardrails
- Content moderation on model output
- PII detection in outputs
- Factual claim flagging for high-stakes domains
- Human review queue for flagged outputs
```

Dimension 5: Accountability — The Accountability Matrix

Scenario	Required Oversight Level
AI suggests a product recommendation	None — user can ignore
AI drafts an email	User reviews before sending
AI updates a database record	Human approval required
AI makes a hiring or lending decision	Human makes final decision, AI only assists
AI takes any irreversible action	Explicit human confirmation every time

Dimension 6: Reliability & Robustness

Key requirements: consistency across runs and user segments, failure handling (API down, malformed response, context window exceeded), graceful degradation path for every AI feature.

Lesson 7.3 — The EU AI Act: PM Summary

Henry speaking:

Risk Level	Examples	Requirements
Unacceptable (Banned)	Social scoring, real-time biometric surveillance	Do not build. Full stop.

High Risk	AI in hiring, credit scoring, medical devices	Mandatory human oversight, transparency, accuracy requirements, EU registration
Limited Risk	Chatbots, AI-generated content	Disclose AI interaction, label AI-generated content
Minimal Risk	Spam filters, recommendation systems	No specific requirements

Lesson 7.4 — The Pre-Launch Ethics Review Template

AI FEATURE ETHICS REVIEW

SECTION 1: HARM ASSESSMENT

1. What is the worst-case harm this feature could cause?
2. Probability: Very Low / Low / Medium / High
3. Severity: Minimal / Moderate / Severe / Critical
4. Risk Score = Probability × Severity

SECTION 2: FAIRNESS ASSESSMENT

5. Which user groups does this feature serve?
6. Have you tested quality across all groups? Results?
7. Are there groups who could be disproportionately harmed?

SECTION 3: TRANSPARENCY ASSESSMENT

8. Do users know they're interacting with AI? How?
9. For AI-assisted decisions: can users understand why?
10. Is there a human appeal/override mechanism?

SECTION 4: GUARDRAILS ASSESSMENT

11. What input guardrails are in place?
12. What output guardrails are in place?
13. What is the human oversight mechanism?
14. What is the incident response plan?

SECTION 5: LAUNCH DECISION

☐ APPROVED ☐ APPROVED WITH CONDITIONS ☐ DELAYED ☐ REJECTED

Signatures: PM ____ | Legal ____ | Trust & Safety ____

Lesson 7.5 — 5 Cultural Practices of Responsible AI Teams

- **Pre-Mortems:** "It's 6 months from now and this feature caused a major harm incident. What happened?"
- **Diverse Red Teams:** Include people with different demographics, backgrounds, and lived experiences.
- **User Feedback Loops for Harm:** Build explicit "Report a problem with this AI response" flows.
- **Incident Post-Mortems:** Blameless post-mortems when AI causes harm. Share learnings across the team.
- **Ethics as a Launch Criterion:** No ethics review = no launch. This must come from PM leadership.

Module 7 Lab Assignment

- Run a Harm Assessment — identify top 3 harms, score probability × severity
- Complete the Privacy Checklist — map every data flow
- Design your 3 Lines of Defense (input guardrails, output guardrails, human oversight)
- Classify your feature under the EU AI Act
- Run the full Pre-Launch Ethics Review and get sign-off from one non-PM stakeholder

Module 8: Lead AI Teams, Manage Stakeholders & Drive Adoption

Lesson 8.1 — The AI PM's Unique Leadership Challenges

Henry speaking:

Challenge 1: Probabilistic Outcomes in a Deterministic Culture

Always anchor AI performance to a human or legacy system baseline. "Our human agents answer correctly 79% of the time. Our AI answers correctly 87% of the time." Never present AI metrics in isolation.

Challenge 2: The Expertise Gap — The "Informed Challenger" Model

You don't need to know HOW to build the model. You need to ask the right questions: "What's the failure mode of this approach?" / "What would it take to get from 87% to 92% accuracy?" / "Is there a simpler approach that gets us to 80% faster?"

Challenge 3: Shifting Ground

Build for model agnosticism from day one. Include in your PRD: "The system must be able to switch between model providers with no more than [X] days of engineering work."

Challenge 4: The "AI Will Do Everything" Expectation

Set expectations before demos. Show a failure case proactively — it builds more trust than hiding failures.

Challenge 5: The Evaluation Communication Gap — Translation Framework

Technical Metric	Business Translation
Faithfulness: 0.87	"13% of responses contain information not in our knowledge base"
Task Completion Rate: 78%	"22% of users don't complete their intended task — 220 users/day needing human support"
p95 Latency: 4.2s	"1 in 20 users waits more than 4 seconds — target is under 3 seconds"

Lesson 8.2 — Building Your AI Team

Henry speaking:

Role	Owns
AI PM (You)	

	Problem definition, success metrics, evaluation strategy, risk register, stakeholder communication, ethics review, roadmap
ML Engineer	Model selection, fine-tuning, evaluation infrastructure
AI/UX Designer	Interaction patterns, trust signals, error states
Data Engineer	Data pipeline, quality, labeling, Golden Dataset
Prompt Engineer	System prompt development, iteration, evaluation

Lesson 8.3 — The AI Roadmap: Horizon Model

Rohan speaking:

Horizon	Timeframe	Criteria	Goal
NOW	0–3 months	High feasibility + High pain	Ship something real. Build credibility.
NEXT	3–6 months	Medium feasibility + High impact	Build the product that drives retention.
LATER	6–12 months	Lower feasibility + Highest strategic value	Build the competitive advantage that's hard to copy.

AI Feature Prioritization Score

Priority Score = (Business Impact × 0.30) + (Feasibility × 0.25) + (User Pain × 0.25) + (Strategic Value × 0.20)

Lesson 8.4 — The 4 Stakeholder Archetypes

Henry speaking:

Archetype	Profile	How to Win Them
The Skeptic	"AI is just hype. Show me the ROI."	Lead with baseline comparison. Hard numbers. Working prototype with real metrics.
The Enthusiast	"AI can do everything!"	Channel their energy. "Here's what we can achieve in 90 days, and here's the roadmap to your bigger vision."
The Risk Manager	"What if it goes wrong?"	Show your risk register. "Here are the 5 failure modes and our mitigation for each."
The End User Champion	"Will this actually help users?"	Involve them in user research. Show real user feedback from pilots.

Stakeholder Communication Cadence

Frequency	Audience	Content
Weekly	Internal team	AI Health Report (3 bullets: improved, regressed, plan)

Monthly	Leadership	Business metrics, quality trend, roadmap progress, top risk
Quarterly	Executive/ Board	ROI summary, competitive positioning, strategic roadmap, compliance status

Lesson 8.5 — The 5 Barriers to AI Adoption

Rohan speaking:

Barrier	How to Overcome It
Trust Deficit	Show sources, show confidence levels, make it easy to verify, start with low-stakes use cases
Workflow Disruption	Meet users where they are, make AI opt-in, show time savings explicitly
Black Box Problem	Explain reasoning, show inputs, make uncertainty visible
Fear of Replacement	Frame as augmentation, involve users in design, show how AI makes them look better
Learning Curve	Provide prompt templates, show examples of great outputs, build progressive disclosure

Lesson 8.6 — AI Adoption Metrics

Henry speaking:

Metric	What It Measures	Target
AI Acceptance Rate	% of AI suggestions users accept without editing	>60%
AI Override Rate	% of AI outputs users significantly edit or reject	<30%
AI Return Rate	% of users who use AI feature again after first use	>50%
Time-to-Value	Time from first AI use to first "aha moment"	<5 minutes

Adoption Health Score = (Acceptance Rate × 0.30) + (Return Rate × 0.25) + (Task Completion Rate × 0.25) + (User Satisfaction × 0.20). Green: >75 | Yellow: 50–75 | Red: <50

Lesson 8.7 — The 5-Phase AI Launch Playbook

Phase	Timing	Key Actions
Seed	Weeks 1–2	10–20 power users, early access, qualitative feedback, case studies
Expand	Weeks 3–4	10% rollout, daily metric monitoring, weekly office hours
Educate	Weeks 5–6	Getting Started guide, 3-min demo video, Tips & Tricks email
Normalize	Weeks 7–8	Full rollout, AI becomes default, celebrate adoption milestones
Optimize	Ongoing	Monthly adoption review, quarterly user interviews, continuous prompt iteration

Module 8 Lab Assignment

- Map your stakeholders using the 4 archetypes — write a 3-sentence communication plan for each
- Build your AI Roadmap using the Horizon Model — score 5 candidate features
- Design your Adoption Strategy — identify top 2 barriers and mitigation plans
- Build your Stakeholder Communication Pack (weekly, monthly, quarterly templates)
- Define your Adoption Health Score with targets for all 5 metrics

Module 9: Build Your AI PM Portfolio & Land the Role

Lesson 9.1 — The AI PM Job Market Reality

Henry speaking:

Demand for AI PMs is growing 40% year-over-year. Most PMs have surface-level AI knowledge. Genuinely qualified AI PMs are scarce. This gap is your opportunity.

The 3 Profiles That Get Hired

Profile	Description	Key Differentiator
The Builder	Has shipped AI products. Has a portfolio. Can talk about failures as fluently as successes.	Commands a premium
The Translator	Deep domain expertise (healthcare, finance, legal) + AI knowledge.	Identifies AI opportunities generalists miss
The Evaluator	Obsessed with quality and measurement. Has built eval suites. Understands LLM-as-Judge.	Rare and highly valued

Lesson 9.2 — The 5-Artifact AI PM Portfolio

Rohan speaking:

Artifact 1: AI Opportunity Assessment (2 pages)

Demonstrates: Problem identification, business thinking, AI suitability judgment.

Include: The Problem (who, frequency, severity, workaround, cost) | The AI Opportunity (why AI, what type, AI Suitability Score) | The Business Case (impact, build cost, run cost, ROI timeline) | The Risk Register (top 3 risks with probability × severity and mitigation)

Artifact 2: The AI PRD (4–6 pages)

Demonstrates: Requirements writing, AI-specific thinking, cross-functional communication.

Must include all AI-specific sections: AI Approach, Data Requirements, Model Selection Rationale, Prompt Strategy, Evaluation Plan, Guardrails Specification, Cost Model, AI Risk Register, Responsible AI Review.

Artifact 3: The Eval Suite

Demonstrates: Quality obsession, technical depth, measurement discipline.

Minimum viable: 30-example Golden Dataset (20 happy path, 7 edge, 3 adversarial) | Custom judge prompt | Scoring spreadsheet | Failure analysis (top 3 patterns) | Improvement log (v1 → v2 → v3)

Artifact 4: Agent Architecture Document

Demonstrates: Systems thinking, agentic AI understanding, technical communication.

Include: Agent Overview (goal, planning pattern, justification) | Agent Stack (Sense/Plan/Act layers) | Tool Schemas | State Object (JSON schema) | Termination Conditions | Failure Mode Analysis | Reliability Checklist

Artifact 5: STAR-AI Case Study

Demonstrates: Storytelling, impact measurement, learning mindset.

SITUATION: Business context, problem, baseline (before AI)
TASK: Your specific role, responsibilities, constraints
ACTION (AI-Specific): AI approach chosen and why, eval criteria, prompt iterations, failure modes encountered, responsible AI considerations
RESULT: Key metric change, cost/latency impact, learnings, what you'd do differently
AI ADDITIONS: Eval scores before/after, cost model at scale, failure rate, next steps

Lesson 9.3 — The 3-Platform Portfolio Strategy

Rohan speaking:

Platform	What to Publish	Goal
LinkedIn	1 post per artifact (5 total), learnings not just outputs, #AIProductManagement	Visibility and inbound opportunities
GitHub	Eval suite code, Golden Dataset (anonymized), prompt templates	Technical credibility signal
Personal Site / Notion	All 5 artifacts linked, AI PM Philosophy statement, contact info	Single URL to share with recruiters

Lesson 9.4 — The AI PM Interview Playbook

Henry speaking:

Type 1: The AI Opportunity Question — "How would you use AI to improve [product]?"

Structure: (1) Identify highest-pain user workflow (2) Apply AI Suitability Test (3) Define specific AI approach (4) Define success metrics (5) Identify top risk. Total: 2.5 minutes. Structured. Specific.

Type 2: The Technical Depth Question — "Walk me through how RAG works."

Structure: (1) Plain English explanation (2) When to use it vs. alternatives (3) The 4 quality levers (4) When RAG fails (5) Specific example from your portfolio

Type 3: The Metrics Question — "How would you measure success of an AI feature?"

Use the 4-Layer KPI Framework: Layer 1 (Model): Faithfulness, relevance, task completion | Layer 2 (Agent): Tool call success, step completion | Layer 3 (Product): Feature adoption, override rate, time-to-value | Layer 4 (Business): Revenue impact, cost savings, NPS delta

Type 4: The Failure Question — "Tell me about an AI project that didn't go as planned."

Structure: (1) What you built and expected (2) What actually happened — be specific (3) Root cause diagnosis (4) What you changed (5) What you'd do differently from day one

Type 5: The Responsible AI Question

Structure: (1) Ethics as product quality, not compliance (2) The 6 dimensions (3) Pre-launch ethics review process (4) Specific ethical trade-off you navigated (5) Your personal philosophy

Lesson 9.5 — AI PM Salary Landscape (2024–2025)

Rohan speaking:

Level	Company Type	Base Salary	Total Comp
PM II (Mid)	Big Tech	\$160K–\$200K	\$250K–\$350K
PM II (Mid)	AI Startup (Series B+)	\$140K–\$180K	\$200K–\$400K (with equity)
Senior PM	Big Tech	\$200K–\$250K	\$350K–\$550K
Senior PM	AI Startup	\$170K–\$220K	\$300K–\$700K (with equity)
Principal/Staff PM	Big Tech	\$250K–\$320K	\$500K–\$900K